

# Design and Development of a Hydrodynamic Tunnel for Experimental Fluid Dynamics

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## Introduction

Hydrodynamic tunnels (HDTs) circulate water to study external flow around immersed bodies under controlled, repeatable conditions – vortex shedding, flow separation and boundary-layer transition. Because water has high density and low kinematic viscosity, target Reynolds numbers are reached at low velocity, making compact lab-scale facilities viable. Sri Lanka has no purpose-built HDT, forcing costly overseas testing and leaving local CFD-validation needs unmet.

## Background

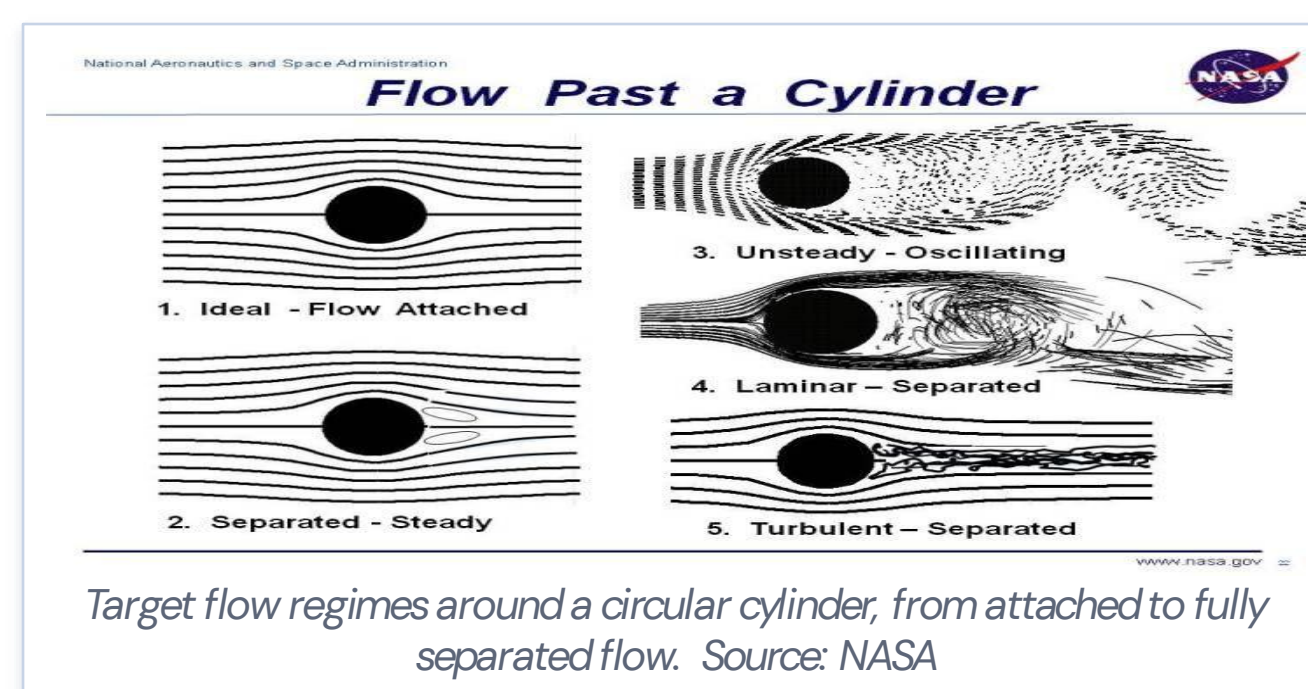
A review of small / medium-scale tunnels examined circuit configurations, contraction and diffuser design, flow conditioning (honeycombs and screens) and head-loss methods. No consolidated, replicable design framework exists for facilities built under real budget, space and material constraints.

## Aim

To design and develop a hydrodynamic tunnel that delivers standard testing flow conditions in the test section, together with effective flow-visualisation methods for experimental fluid dynamics.

## Objectives

1. Identify relationships between tunnel design characteristics and test-section flow quality.
2. Design the tunnel based on the analysis of these design characteristics.
3. Optimise the design (CFD + FEA) to ensure standard flow conditions.
4. Develop and validate the tunnel to achieve standard flow in the test section.



## References

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## Materials & Methods

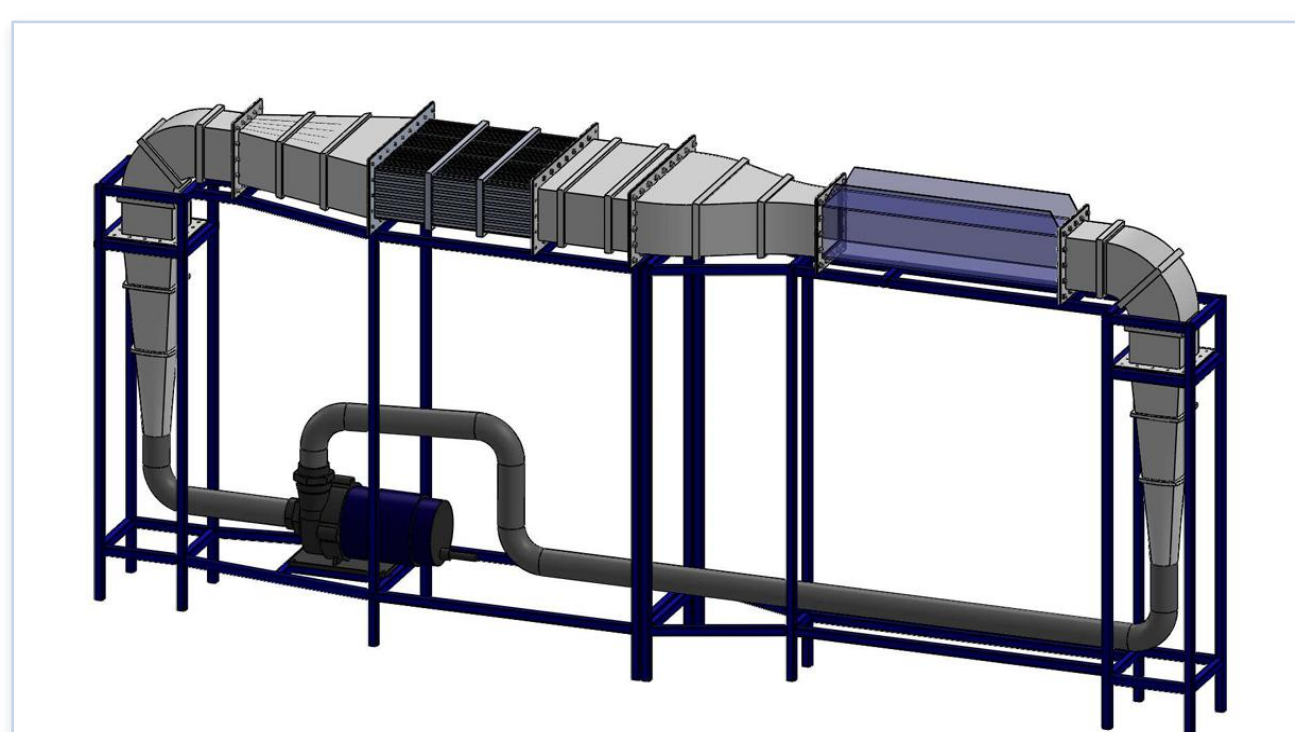
A four-phase process led from literature-based configuration selection to a fabricated tunnel. A vertical, closed-loop circuit with an open-channel test section was selected against budget, height and Reynolds-number constraints. Components were sized analytically; a MATLAB head-loss model verified pump adequacy. Single-phase ANSYS Fluent CFD and FEA optimised flow and structure. Primary parts were fabricated in glass-fibre (GFRP) by hand layup on glass moulds.



**150 × 150**  
Test section (mm)

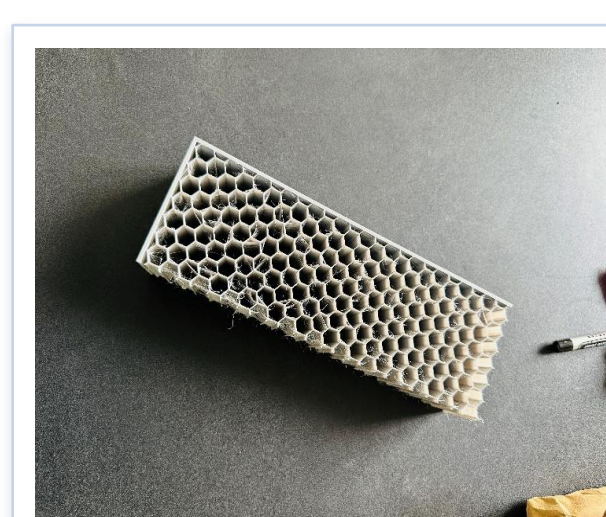
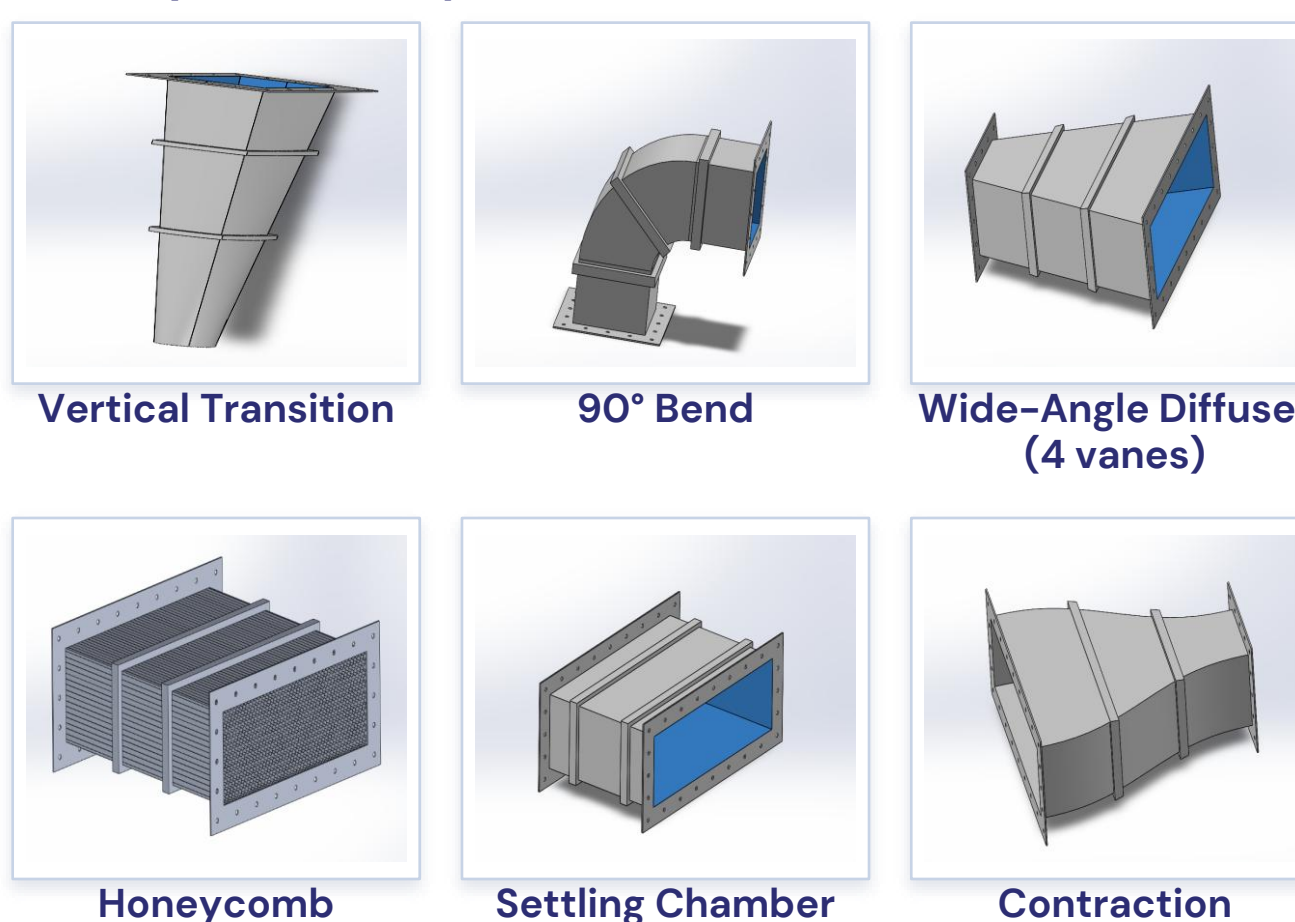
**0.0136**  
Flow rate Q (m³/s)

**0 – 8,000**  
Reynolds number, Re

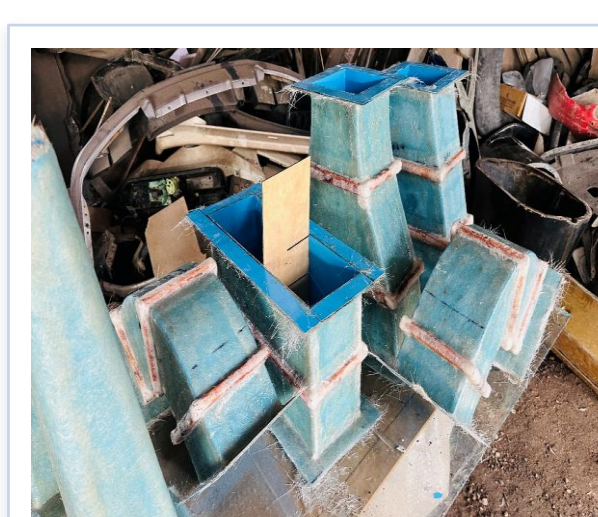


Complete CAD model of the vertical closed-loop hydrodynamic tunnel (Group 14).

### Primary GFRP components



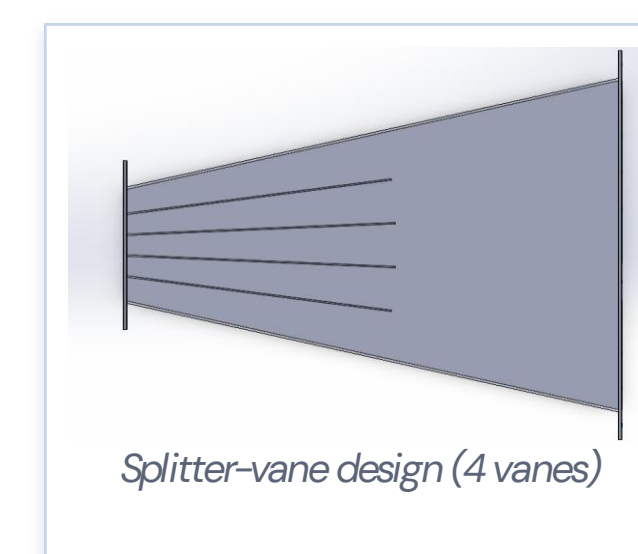
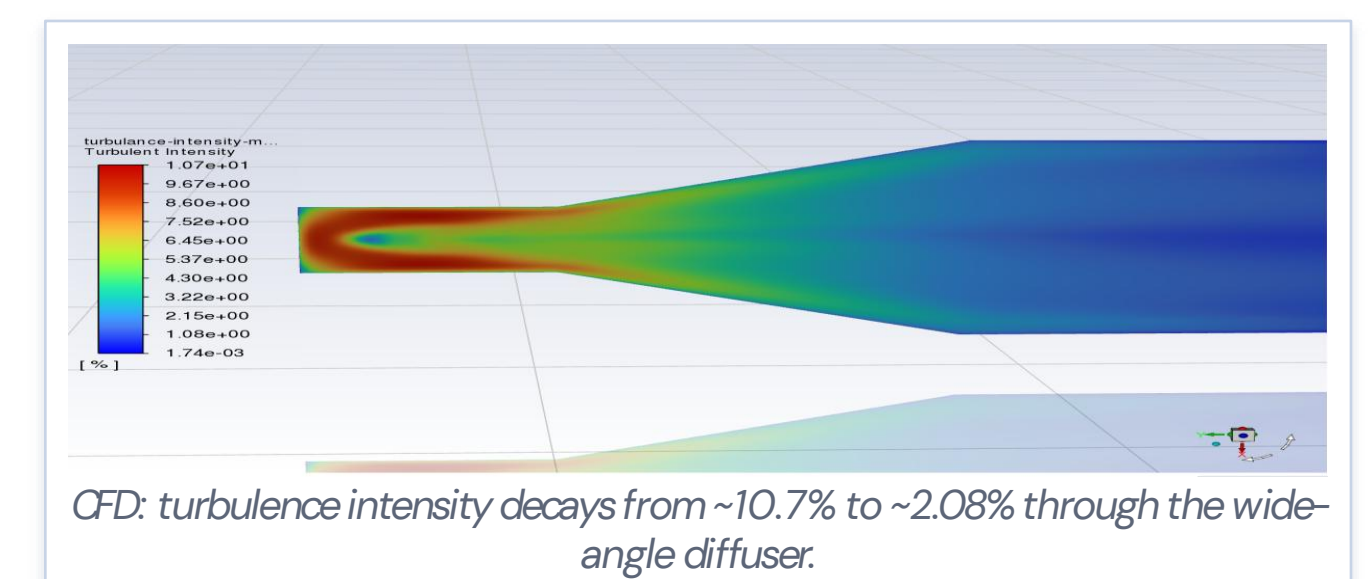
3D-printed PLA honeycomb flow straightener



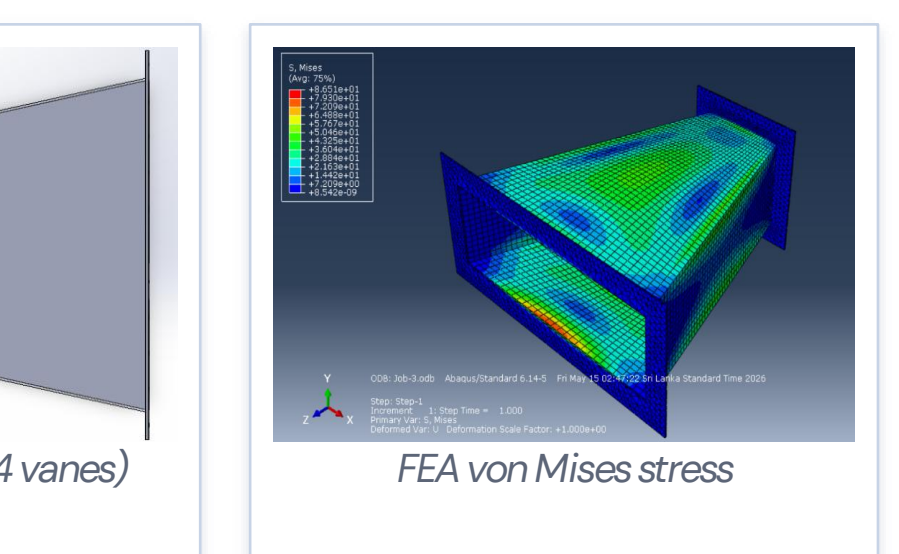
Hand-laid GFRP components

## Results & Validation

Transient CFD of the 35° wide-angle diffuser revealed severe flow separation; four guide vanes (7° per channel, after Waitman–Reneau–Kline) were designed to suppress it. FEA showed the unreinforced 3 mm GFRP wall exceeds the 60 MPa allowable, so external ribs and extra GFRP layers were added.



**0.593**  
Pressure recovery,  $C_p$



**2.08 %**  
Turbulence intensity

**≈ 0.9 m**  
Head loss (pump 9.14 m)



Fully assembled & commissioned tunnel



Dye visualisation around a cylinder

## Conclusions & Future Works

- Sri Lanka's first hydrodynamic tunnel was designed, fabricated and commissioned, with a consolidated, replicable design framework.
- Low-cost methods validated: GFRP hand layup on glass moulds (smooth inner finish) and a 3D-printed PLA honeycomb.
- Future: install splitter vanes; commission hydrogen-bubble visualisation; add a VFD for variable-speed Re sweeps; fit flanged modular pipe couplings; upgrade to a precision honeycomb.

## Acknowledgement

We sincerely thank our supervisors Dr. I. D. Nissanka, Dr. N. Samaraweera and Mr. L. de Silva, and the academic and non-academic staff of the Department of Mechanical Engineering, University of Moratuwa, for their guidance and support.